

Draft for an Academic Address

Having discharged the office of public professor for more than twenty years, and having experienced throughout this long time the truth of the proverb that by teaching we learn, it does not seem inappropriate to me, when the occasion is given, to say a few things about the true way of cultivating analytical science. If I wished to pursue this subject with the fullness it deserves, neither time nor strength would suffice. It is enough to warn by a few examples of the danger one is in if, in matters which one has scarcely looked at, much less penetrated, one presumes to establish a fixed order. It is agreed by all that many geometers, generally following the chronological origin of the individual parts of doctrine, have established a certain order of topics in which one must ascend from each part to the next higher one. They have thought that this order must be observed so strictly that one not rarely falls into assertions such as these: in treating the calculus of partial differentials one must suppose that the integration of ordinary differential equations has already been completely perfected; and similarly, if a question belonging to higher analysis is proposed, the solution of algebraic equations must be ready to hand. How greatly nature itself mocks this artificial order of topics and difficulties, and how many aids would be lacking if one wished to preserve such an order in analytical investigations, I now undertake to show by a few examples.

What else is EULER's celebrated invention concerning the addition of integrals, which we now call elliptic, than a fundamental theorem concerning these integrals, drawn from a higher chapter? It is well known that this great man derived this distinguished truth from the consideration of a differential equation whose variables are separated. And who will doubt that the addition theorem is the true foundation of the entire doctrine of elliptic functions, if he considers not only that nearly all of the discoveries of the illustrious LEGENDRE in this chapter rest upon Euler's theorem, but also that on the same foundation rest the excellent truths by which the young ABEL and JACOBI, deserving their first laurels and contending without envy, enlarged this most fertile part of doctrine, and moved the admiration of that Nestor of geometers? In commemorating these brilliant discoveries there is renewed for me the grief which I share with all who care for the growth of analysis: inexorable fate took from the republic of letters two such great minds by premature death, one scarcely having entered the arena, the other diverted by so many and such great investigations that he was not allowed to put the final hand to them.

Another example of aid sought from parts of doctrine that seem to surpass the proposed question in difficulty is provided by the celebrated JACOBI's investigations on shortest lines on the surface of an ellipsoid. The problem is easily reduced to a problem of mechanics by means of the known principle that a point constrained by no accelerating force on a given surface moves with its initial velocity along a shortest line. Since the dynamical equations on which the motion on surfaces of the second degree depends do not admit integration by known methods, that illustrious man ascended to a higher region and, using a theorem of the illustrious HAMILTON, which he had previously reduced to a simpler form, reduced the question to a single equation containing partial differentials and was thereby able to solve it with excellent success. By this ingenious invention he founded an entirely new chapter in the theory of surfaces of the second order, in the cultivation of which the acute men JOACHIMSTHAL, CHASLES, LIOUVILLE, and ROBERTS found ample material for exercising their powers and enriching geometry with distinguished truths.

I now come to an argument than which one can find nothing more admirable, if one wishes to show by example how often the parts of analytical science are bound together by links of art, though at first sight hardly any connection is seen among them. I speak of the geometrical investigations concerning the section of the circle, a question known to have been discussed already in the time of EUCLID. The solution of the problem in the

simplest cases - the only ones that seem able to dispense with the calculus of the moderns - having been completed by the Alexandrians, the general question awaited the efforts of moderns. Among them the great FRANÇOIS VIÈTE first succeeded in reducing the general division of the circle to an algebraic equation. After the interval of a century, the equation defining the division of the circle was reduced to the utmost degree of simplicity by the work of the illustrious COTES and MOIVRE, who showed that the problem depends on the simplest of all equations, in which the power of the unknown is equal to unity. After another century the sagacious VANDERMONDE undertook most ingenious investigations on the Cotesian equation. We now see that they rest on a very profound principle, though they seem scarcely to have come to the knowledge of contemporary geometers, perhaps because the author, already advanced in age when he first gave his attention to analysis, set them out somewhat obscurely. I shall briefly recount the very singular origin of these investigations, which the illustrious LAGRANGE rescued from oblivion in the notes added to his treatise on numerical solution of equations, as my revered teacher LACROIX, a man highly meritorious in the diffusion of analytical doctrine, once told it to me. VANDERMONDE, Dutch by nation, after practicing medicine for many years in his country and then living in Paris at an advanced age, frequented a certain establishment where many gathered for games. Seeing that the famous geometer CLAIRAUT was treated with singular reverence by everyone who came there, he, entirely untrained in analysis, asked someone whether anything remained to be perfected in analytical doctrine. He received the reply that, among other things - and we may add, many things - the general solution of algebraic equations was still lacking. Stirred by this reply, the acute man mastered the elements of algebra in a short time, devoted himself with the greatest zeal to this subject, and soon completed the investigations mentioned above.

At last appeared the illustrious GAUSS's *Disquisitiones arithmeticae*, a work by which its author inaugurated a century and restored to Germany the glory which, after LEIBNIZ's time, had belonged to foreigners, at a time when the cultivation of analysis among us was languishing and almost extinct. In the last section of the work, to pass over other things, the equation belonging to the section of the circle is solved generally - or rather, to define the matter more clearly, the equation is reduced to pure equations of lower degrees. By this reduction the section of the circle is carried out geometrically in infinitely many cases, the simplest after the previously known cases being the division into seventeen parts, a discovery that soon came to be on the lips even of beginners, while the other chapters of the work, no less worthy of admiration, scarcely became known to a few foreign geometers. The author himself is witness to this, for more than twenty years after the book was published he publicly declared that he had still found no one who had read it through. But I return to the matter.

The solution is built on a very singular principle taken from higher arithmetic. By its aid the roots of the equation are arranged in a cycle in such a way that each depends on the preceding in the same manner; once this is done the reduction to lower equations is accomplished without difficulty. Yet in solving this algebraic problem the great man did not rest content: looking more deeply into the proposed subject, with his characteristic sagacity he soon perceived that the solution was most closely connected with the most hidden properties of numbers. Algebra, returning with great interest the help it had borrowed from higher arithmetic, drew from this rich source most beautiful arithmetical theorems, which scarcely seem accessible by any other path. Following in the footsteps of this great man, the distinguished geometers ABEL, JACOBI, CAUCHY, GALOIS, EISENSTEIN, KUMMER, and LEBESGUE formed from this subject a very extensive discipline, which one can scarcely say should be referred more to algebra or to higher arithmetic. The boundaries of the two are so confounded that they cannot be discerned. To the same subject belong the observations of the celebrated POINSOT on primitive roots derived from the section of the

circle. How this geometer could assert that by these observations he had presented the first application of algebra to the knowledge of numerical properties, I could hardly understand, unless I suppose that this acute man once bore some resemblance to the good Homer portrayed by HORACE in the *Ars poetica*.

I could bring other examples to show that great aids for analytical questions can be sought from regions apparently most remote, if I wished to speak of the use of integral calculus and of series with an infinite number of terms in higher arithmetic; but after the brilliant discoveries just recalled I fear it would not be fitting to discuss my own.

If I may comprehend in a few words all that I have said above and express it under the form of a trope, those very great studies and contests, to which through so many centuries men endowed with supreme genius and singular sagacity have devoted their labor, do not seem to me very different from the composition of a certain singular book, whose chapters, pages, and lines all of us, each according to his powers, try to compose. Since in that book so many passages are left to posterity to perfect and so many pages to fill in, no certain order seems to be established, nor will anyone presume so to join individual sentences together that Horace's rule, "let not the middle disagree with the beginning, nor the end with the middle," can rightly be applied.

To this book, then, there seem rightly to belong the words of a man of outstanding acuteness of mind and much meditation on the human mind:

The last thing one finds in making a work is to know what must be put first.